

Ferrite Core Selection Using the K_g Method

Input Parameters

$L = 34.9 \times 10^{-6} \text{ H}$	(Inductance)
$I_o = 4.583 \text{ A}$	(DC Current)
$\Delta I = 1.375 \text{ A}$	(AC Ripple Current)
$P_o = 110 \text{ W}$	(Output Power)
$\alpha = 1\%$	(Regulation)
$f_s = 100 \times 10^3 \text{ Hz}$	(Switching Frequency)
$B_m = 0.25 \text{ T}$	(Operating Flux Density)
$K_u = 0.4$	(Window Utilization)
$T_r = 25^\circ \text{C}$	(Temperature Rise Goal)

Calculations

1. Peak Current (I_{pk}):

$$I_{pk} = I_o + \frac{\Delta I}{2} = 4.583 + \frac{1.375}{2} = 5.27 \text{ A}$$

2. Energy-Handling Capability (E):

$$E = \frac{1}{2} L I_{pk}^2 = \frac{1}{2} \cdot 34.9 \times 10^{-6} \cdot (5.27)^2 = 4.87 \times 10^{-4} \text{ J}$$

3. Electrical Condition Coefficient (K_e):

$$K_e = 0.145 \cdot P_o \cdot B_m^2 \cdot 10^{-4} = 0.145 \cdot 110 \cdot (0.25)^2 \cdot 10^{-4} = 9.96 \times 10^{-5}$$

4. Core Geometry Coefficient (K_g):

$$K_g = \frac{E^2}{K_e \cdot \alpha} = \frac{(4.87 \times 10^{-4})^2}{9.96 \times 10^{-5} \cdot 1} = 2.38 \times 10^{-3}$$

Notes

- K_g determines the minimum core size required
- Select a ferrite core with $K_g \geq 2.38 \times 10^{-3} \text{ cm}^5$
- Verify core saturation at $B_m = 0.25 \text{ T}$